

# Abdellah Fourtassi

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Laboratoire d'Informatique et des Systèmes  
Campus universitaire de Luminy  
Aix Marseille Université

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Team website: [cocodev.fr](http://cocodev.fr)

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**Research Profile.** I combine theoretical and practical tools from computer science to build formal models of human cognition. I use machine learning as a model of cognitive development, with a particular focus on first language acquisition. I am especially interested in quantifying the role of multimodal and social grounding in learning.

## APPOINTMENT

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### Aix Marseille University

Maître de Conférences HDR (~ Associate Professor), Computer Science, 2025–present  
Maître de Conférences (~ Assistant Professor), Computer Science, 2019–2025  
Researcher, Natural Language Processing research group (TALEP), 2019–present  
Research fellow, Institute of Language, Communication & the Brain (ILCB), 2019–present  
Co-Director, Master's program of Cognitive Science (MASCO), 2020–2021

### Institut National de Recherche en Sciences et Technologies du numérique (INRIA, Paris)

Visiting Researcher, *Délégation Recherche INRIA*, 2021–2022

### Stanford University

Postdoctoral Research Fellow, Department of Psychology, 2016–2019

## EDUCATION

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### Ecole Normale Supérieure (ENS), Paris

Ph.D. in Cognitive Science, *summa cum laude*, 2015

### Ecole des Hautes Etudes en Sciences Sociales (EHESS), Paris

Master's degree in Cognitive Science (CogMaster), 2012

### Ecole Polytechnique, Paris

Science & Engineering diploma, 2012

### Lycée Moulay Youssef, Rabat

Undergraduate studies in Mathematics & Physics, CPGE/MP\*, 2006–2008

## **DISTINCTIONS, AWARDS & SELECTED ACHIEVEMENTS**

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Keynote Speaker, *Child Language Symposium (CLS)*, 2026

Keynote Speaker, *SIGDIAL*, 2025

Aix Marseille University Best PhD Award, 2024 (thesis by M. Nikolaus; co-supervised with L. Prévot)

Marie Skłodowska-Curie European Fellowship, €185k, 2019 (declined; tenured position at AMU)

Stanford Data Science Seed Grant Award, \$50k, 2018

Fyssen Foundation's postdoctoral Study Grant, €40k, 2016

Visiting Scholar, Stanford University, March–July, 2016 (funded research position).

France-Stanford Center for Interdisciplinary Studies' Collaborative Project Award (\$15k), 2015–2016

Finalist, CoNLL Best Paper Award – *Fourtassi & Dupoux (2014)*

ACL Best Student Paper Award (\$1,000) – *Fourtassi & Dupoux (2013)*

Ranked 2<sup>nd</sup>/30, CMCL shared task on Eye-Tracking Prediction – *Dary, Nasr, & Fourtassi (2021)*

Doctoral Fellowship (AMX — Allocation-Moniteur Polytechnicien), 2012–2015

Eiffel Excellence Scholarship (undergraduate) — French Ministry of Foreign Affairs, 2008–2012

Early distinctions: Admission to École Polytechnique via competitive examination; ranked 12th nationwide in Morocco's national engineering-school entrance examination.

## **RESEARCH GRANTS**

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ANR JCJC (France), €287k (PI), 2022–2025

*Mastering the Art of Conversation in Middle Childhood (MACoMiC)*

ESRC (UK), £300k (co-PI, PI: G. Vigliocco, UCL), 2024–2027

*Language Acquisition in the Early Years: Learning From and With Caregivers*

Leverhulme Trust (UK), £290k (co-PI, PI: G. Vigliocco, UCL), 2025–2028,

*What Makes Social Interaction a Powerful Learning Tool*

## **PEER-REVIEWED PUBLICATIONS**

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### **Book**

**Fourtassi, A.** (under contract). *Neural Networks for a Theory of Language Development*.  
Cambridge University Press

## Edited Special Issue

Agee, L., Monfils, M, **Fourtassi, A.** (2025). Editorial: The Neural and Behavioral Methods of Social Learning. *Frontiers in Behavioral Neuroscience*

Apidianaki, M., **Fourtassi, A.**, and Padó, S. (2024). Editorial: Language Learning, Representation, and Processing in Humans and Machines. *Computational Linguistics*

## Journal papers

§ indicates student authors under my supervision

1. §Agrawal, A., Favre, B., & **Fourtassi, A.** (under revision). Mapping the Communicative Landscape of Early Child-Caregiver Dialogue.
2. **Fourtassi, A.**, & §Xu, Z. (under review). Initiating vs. Responding to Questions in Child-caregiver Spontaneous Conversations Across Preschool.
3. Nikolaus, M., & **Fourtassi, A.** (2026). Modeling Children's Grammar Learning via Caregiver Feedback in Natural Conversations. *Philosophical Transactions of the Royal Society B*, 381.
4. Bodur, K., **Fourtassi, A.**, Fredouille, C., Meunier, C. (2025). Speech Reduction in Middle Childhood: the Case of Child-Caregiver Interactions. *First Language*
5. §Mazzocconi, C., O'Brien, B., §Bodur, K., & **Fourtassi, A.** (2025). Do Children Laugh Like Their Parents? Conversational Laughter Mimicry Occurrence and Acoustic Alignment in Middle-Childhood. *Journal of Nonverbal Behavior*, 49.
6. Agee, L., **Fourtassi, A.**, and Monfils, M. (2025). Social Context as a Source of Variability in the Psychological Sciences. *Frontiers in Behavioral Neuroscience*, 18.
7. §Cabiddu, F., §Nikolaus, M., & **Fourtassi, A.** (2025). Comparing Children and Large Language Models in Word Sense Disambiguation: Insights and Challenges. *Language Development Research*, 5(1).
8. §Bodur, K., §Nikolaus, N., Prévot, L., & **Fourtassi, A.** (2023). Using video calls to study children's conversational development: The case of backchannel signaling. *Frontiers in Computer Science*, 5.
9. §Wilson, K., Frank, M., C., & **Fourtassi, A.** (2023). Conceptual Hierarchy in Child-Directed Speech: Implicit Cues are More Reliable. *Journal of Cognition & Development*, 24(4).

10. §Nikolaus, M. & **Fourtassi, A.** (2023). Communicative Feedback in Language Acquisition. *New Ideas in Psychology*, 68, 100985.
11. §Nikolaus, M., §Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2022). Large-scale Study of Speech Acts' Development in Early Childhood. *Language Development Research*, 2(1).
12. §Jiang, H., Frank, M., C., Kulkarni, V., & **Fourtassi, A.** (2022). Exploring Patterns of Stability and Change in Caregivers' Word Usage Across Early Childhood. *Cognitive Science*, 46 (7).
13. **Fourtassi, A.**, §Regan, S., & Frank, M., C. (2021). Continuous Developmental Change Explains Discontinuities in Word Learning. *Developmental Science*, 24(2).
14. **Fourtassi, A.** & Frank, M. C. (2020) How Optimal is Word Identification Under Multimodal Uncertainty? *Cognition*, 199.
15. **Fourtassi, A.**, §Bian, Y. & Frank, M. C (2020). The Growth of Children's Semantic and Phonological Networks: Insight from 10 Languages. *Cognitive Science*, 44 (7).

## Peer-reviewed Proceedings

16. Jumelet, J., **Fourtassi, A.**, Haga, A., Bunzeck, B., Shandilya, B., Galvan-Sosa, D., Haznitrana, F. G., Padovani, F., Meyer, F., Hu, H., Etzaniz, J., Prevot, L., He, L., Grandury, M., Marcheva, M., Foroutan, N., Theodoropoulos, N., Sadeghi, P., Song, S., ... Choshen, L. (under review). *BabyBabelLM: A Multilingual Benchmark of Developmentally Plausible Training Data*.
17. §Brahimi, Y., §Blanc, C., & **Fourtassi, A.** (2026). Predicting Turn-Taking in Child-Adult Conversations Using Voice Activity Projection. In **IWSDS** (16th International Workshop on Spoken Dialogue Systems Technology).
18. §Liu, J., & **Fourtassi, A.** (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. In **SemDial** (29th Workshop on the Semantics and Pragmatics of Dialogue)
19. §Agrawal, A., Favre, B., & **Fourtassi, A.** (2025). Identifying Repair Opportunities in Child-Caregiver Interactions. In **SemDial** (29th Workshop on the Semantics and Pragmatics of Dialogue)
20. §Agrawal, A., §Nikolaus, M., Favre, B., & **Fourtassi, A.** (2024). Automatic Coding of Contingency in Child-Caregiver Conversations. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)

21. §Nikolaus, M., §Agrawal, A., §Kaklamanis, P., Warstadt, A., & **Fourtassi, A.** (2024). Automatic Annotation of Grammaticality in Child-Caregiver Conversations. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)
22. §Goumri, D., §Agrawal, A., §Nikolaus, M., §VU, T., Bodur, K., §Semmar, E., §Armand, C., §Mazzocconi, C., Gupta, S., Prévot, L., Favre, B., Becerra, L., & **Fourtassi, A.** (2024). A Developmental Corpus of Child-Caregiver's Face-to-face vs. Video Call Conversations in Middle Childhood. In **LREC-COLING** (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation)
23. §Schweitzer, M., Montagnini, A., **Fourtassi, A.**, & Schatz, T. (2024). Multimodal Integration in Audio-Visual Speech Recognition: How Far Are We From Human-Level Robustness? In **NeurIPS** Workshop on Behavioral Machine Learning.
24. §Goumri, D.-E., Becerra, L., & **Fourtassi, A.** (2024). Child-Caregiver Gaze Dynamics in Naturalistic Face-to-Face Conversations. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
25. §Agrawal, A., Favre, B., & **Fourtassi, A.** (2024). Analyzing Communicative Intent Coordination in Child-Caregiver Interactions. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
26. §Peirola, M., § Xu, Z., & **Fourtassi, A.** (2024). Development of Flexible Role-Taking in Conversations Across Preschool. In **CogSci** (46th Annual Meeting of the Cognitive Science Society).
27. §Goumri, D., Janssoone, T., Becerra, L., & **Fourtassi, A.** (2023). Automatic Detection of Gaze and Smile in Children's Video Calls. In **ICMI Companion** (International Conference on Multimodal Interaction)
28. §Cabiddu, F., §Nikolaus, M., & **Fourtassi, A.** (2023). Modeling Children's Word Sense Disambiguation with Large Language Models: Insights and Challenges. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
29. §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2023). Communicative Feedback Signals in Response to Children's Grammatical Errors. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
30. §Mazzocconi, C., O'Brien, B., El Haddad, K., §Bodur, K., **Fourtassi, A.** (2023). Differences between Mimicking and Non-Mimicking Laughter in Child-Caregiver Conversation: A

Distributional and Acoustic Analysis. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).

31. §Agrawal, A., §Liu, J., §Bodur, K., Favre, B., & **Fourtassi, A.** (2023). Development of Multimodal Turn Coordination in Conversations: Evidence for Adult-like Behavior in Middle Childhood. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
32. Tiran, T., Meewis, F., & **Fourtassi, A.**, & Dautriche I. (2023). Typology of Topological Relations Using Machine Translation. In **CogSci** (45th Annual Meeting of the Cognitive Science Society).
33. §Nikolaus, M., Salin, E., Ayache, S., **Fourtassi, A.**, & Favre, B. (2022). Do Vision-and-Language Transformers Learn Predicate-Noun Dependencies? In **EMNLP** (Conference on Empirical Methods in Natural Language Processing).
34. §Liu, J., §Nikolaus, M., §Bodur, K., & **Fourtassi, A.** (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation. In **ICMI Companion** (International Conference on Multimodal Interaction)
35. §Misiek, T. & **Fourtassi, A.** (2022). Caregivers Exaggerate Their Lexical Alignment to Young Children Across Several Cultures. In **SemDial** (26th Workshop on the Semantics and Pragmatics of Dialogue).
36. §Hallart\*, C., §Peirola\*, M., §Xu, Z., & **Fourtassi, A.** (2022). Development of Dialog Contingency in Child-Caregiver Early Interaction (\* shared first authorship). In **SemDial** (26th Workshop on the Semantics and Pragmatics of Dialogue).
37. §Bodur, K., §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Backchannel Behavior in Child-Caregiver Video Calls. In **CogSci** (44th Annual Meeting of the Cognitive Science Society).
38. §Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Communicative Feedback as a Mechanism Supporting the Production of Intelligible Speech in Early Childhood. In **CogSci** (44th Annual Meeting of the Cognitive Science Society).
39. §Nikolaus, M., & **Fourtassi, A.** (2021). Modeling the Interaction between Perception-based Learning and Production-based Learning in Children's Early Acquisition of Semantic Knowledge. In **CoNLL** (Conference on Computational Natural Language Learning)
40. §Dary, F., Nasr, A., & **Fourtassi, A.** (2021). TALEP at CMCL 2021 Shared Task: Non-Linear Combination of Low and High-level Features for Predicting Eye-Tracking Data. In **CMCL** (11th Workshop on Cognitive Modeling and Computational Linguistics).  
**2nd place (out of 30 submissions)**

41. §Bodur, K., §Nikolaus, M., §Kassim, F., Prévot, L., & **Fourtassi, A.** (2021). ChiCo: A Multimodal Corpus for the Study of Child Conversation. In **ICMI Companion** (International Conference on Multimodal Interaction).
42. §Nikolaus, M. & **Fourtassi, A.** (2021). Evaluating the Acquisition of Semantic Knowledge from Cross-situational Learning in Artificial Neural Networks. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics).
43. §Nikolaus, M., §Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2021). Large-scale Study of Speech Acts' Development Using Automatic Labeling. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
44. §Nikolaus, M., §Maes E., & **Fourtassi, A.** (2021). Modeling Speech Act Development in Early Childhood: The Role of Frequency and Linguistic Cues. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
45. §Dary, F., **Fourtassi, A.**, & Nasr, A., (2021). On the Role of Low-level Linguistic Tasks for Reading Time Prediction. In **CogSci** (43th Annual Meeting of the Cognitive Science Society).
46. §Misiek, T., Favre, B., & **Fourtassi, A.** (2020). Development of Multi-level Linguistic Alignment in Child-Adult Conversations. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics)
47. **Fourtassi, A.** (2020). Word Co-occurrence in Child-directed Speech Predicts Children's Free Word Associations. In **CMCL** (The 11th Workshop on Cognitive Modeling and Computational Linguistics).
48. **Fourtassi, A.**, §Wilson, K., & Frank, M., C. (2020). Discovering Conceptual Hierarchy Through Explicit and Implicit Cues in Child-Directed Speech. In **CogSci** (42nd Annual Meeting of the Cognitive Science Society).
49. **Fourtassi, A.**, §Scheinfeld, I., & Frank, M. C.(2019) The Development of Abstract Concepts in the Children's Early Lexical Networks. In **CMCL** (The 10th Workshop on Cognitive Modeling and Computational Linguistics).
50. **Fourtassi, A.**, §Regan, S. & Frank, M. C. (2019). The Role of Continuous Developmental Change in Explaining Discontinuities in Word Learning. In **CogSci** (41st Annual Meeting of the Cognitive Science Society).
51. **Fourtassi, A.**, & Dupoux, E. (2019) Phoneme Learning is Influenced by the Taxonomic Similarity of the Semantic Referents. In **CogSci** (41st Annual Meeting of the Cognitive Science Society).

52. **Fourtassi, A.**, §Bian, Y., & Frank, M., C. (2018). Word Learning as Network Growth: A Cross-Linguistic Analysis. In **CogSci** (40th Annual Meeting of the Cognitive Science Society).
53. **Fourtassi, A.** & Frank, M., C. (2017). Word Identification Under Multimodal Uncertainty. In **CogSci** (39th Annual Meeting of the Cognitive Science Society).
54. **Fourtassi, A.** & Dupoux, E. (2016). The role of word-word co-occurrence in word meaning learning. In **CogSci** (38th Annual Meeting of the Cognitive Science Society).
55. **Fourtassi, A.** & Dupoux, E. (2014). Rudimentary Lexicon and Semantics Help Bootstrap Phoneme Acquisition. In **CoNLL** (Conference on Computational Natural Language Learning) **Best paper award finalist**
56. **Fourtassi, A.**, Schatz, T., Varadarajan, B. & Dupoux, E. (2014). Exploring the Relative Role of Bottom-Up and Top-Down Information in Phoneme Learning. In **ACL** (52nd Annual Meeting of the Association for Computational Linguistics).
57. **Fourtassi, A.**, Dunbar, E., & Dupoux, E. (2014). Self Consistency as an Inductive Bias in Early Language Acquisition. In **CogSci** (36th Annual Meeting of the Cognitive Science Society).
58. **Fourtassi, A.**, Borschinger, B., Johnson, M. & Dupoux, E. (2013). Why is English so easy to segment? In **CMCL** (The 4th Workshop on Cognitive Modeling and Computational Linguistics).
59. **Fourtassi, A.** & Dupoux, E. (2013). A Corpus-based Evaluation Method for Distributional Semantic Models. In **ACL student research workshop** **IBM Best Student paper (\$1,000)**
60. Jansen, A., Dupoux, E., Goldwater, S., Johnson, M., Khudanpur, S., Church, K., Feldman, N., Hermansky, H., Metze, F., Rose, R., Seltzer, M., Clark, P., McGraw, I., Varadarajan, B., Bennett, E., Borschinger, B., Chiu, J., Dunbar, E., **Fourtassi, A.**, Harwath, D., Lee, C., Levin, K., Norouzian, A., Peddinti, V., Richardson, R., Schatz, T. & Thomas, S. (2013). A summary of the 2012 JHU CLSP workshop on zero resource speech technologies and models of early language acquisition. In **ICASSP** (The International Conference on Acoustics, Speech and Signal Processing)

## INVITED TALKS

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### Keynote Lectures

Child Language Symposium (CLS), *Deep Learning for a Theory of Language Development*, 2026

SIGDIAL, *The Primary Role of Dialogue in Language Development*, 2025

TALN Workshop on AI for Education, *Can Large Language Models Learn Like Children?* 2025



## Invited Talks — Workshops & Expert Meetings

*Machine Learning to Study Home Audio Recordings*, Online Event, International Congress of Infant Studies (ICIS), June 19, 2025 ([link](#))

*State of the Art in Scientific Psychology*, National Meeting, Comité National Français de Psychologie Scientifique (CNFPS), Oct 25, 2025 ([link](#))

*NLP in the Era of Generative AI, Cognitive Sciences, and Societal Transformation*, Workshop, Mila Quebec Artificial Intelligence Institute, Oct 1 – Oct 3, 2024 ([link](#))

*Developmental Machine Learning: From Human Learning to Machines and Back*, Dagstuhl Seminar, Oct 16 – Oct 21, 2022 ([link](#))

*AI and the Productivity of Science*, Workshop, Organisation for Economic Co-operation and Development (OECD), Oct 29 – Nov 5, 2021 ([link](#))

## Invited Talks — Universities & Research Consortia (Selection)

Social Contingency Consortium, 2023, 2025

Cambridge University, Center for Family Research, 2024

University College London, Psychology and Language Sciences, 2023, 2024

International Centre for Language and Communicative Development (LuCiD), 2023

University of Göttingen, Psychology of Language Department, 2023

The Chinese University of Hong Kong, Department of Linguistics, 2022

INRIA Lille Nord Europe, MGNET Seminar, 2022

Radboud University, Multimodal Language and Cognition Seminar, 2022

Pr[AI]rie institute Colloquium, 2022

University of Aarhus, Interacting Minds Center Seminar, 2022

University of Amsterdam, Computational Linguistics Seminar (CSL), 2022

University Grenoble Alpes, GIPSA-lab seminar, 2022

Stanford University, Center for the Study of Language & Information (CSLI), 2018

## MENTORSHIP

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### Ph.D. Supervision — Primary Role

Marianne Schweitzer (2025–present) — co-advised with C. François

*Funding: Amidex fellowship*

Youcef Brahim (2025–present)

*Funding: ED184 fellowship*

Dhia Elhak Goumri (2022–present) — co-advised with L. Becerra

*Funding: MACoMiC grant (ANR JCJC)*

*Defense expected: March 2026*

Abhishek Agrawal (2022–2025) — co-advised with B. Favre

*Funding: MACoMiC grant (ANR JCJC)*

*Defended: October 2025*

Mitja Nikolaus (2020–2023) — co-advised with L. Prévot

*Funding: ILCB fellowship*

*Defended: September 2023; **Aix Marseille University Best PhD Award***

*Current position: Postdoctoral Researcher*

## Ph.D. Supervision — Secondary Role

Mathias Vigouroux (2025–present) — co-advised with J. Sackur

*Funding: ED540*

Franck Dary (2018–2022) — co-advised with A. Nasr

*Funding: ED184 fellowship*

*Defended: July 2022*

*Current position: Data Scientist Tech Lead, DialOnce*

## Postdoctoral Researchers

Giulia Rambelli (2026–present) — co-advised with L. Prévot

*Funding: ILCB Fellowship*

Monica Barbir (2024–present)

*Funding: Marie Skłodowska-Curie European Fellowship*

Chiara Mazzocconi (2020–2023) — co-advised with B. Priego-Valverde

*Funding: ILCB Fellowship*

## Master's Interns

\* Supervision resulted in peer-reviewed publication

Youcef Brahim\* — Image and Signal Processing (M2), Toulouse University, 2025  
 César Blanc\* — Artificial Intelligence (M2), Aix Marseille University, 2025  
 Nina Gregorio — Cognitive Science (M1), Ecole Normale Supérieure, 2023  
 Jing Liu\* — Linguistics & Communication (M2), Radboud University, 2022  
 Maryam Taiaa — Data Science (M2), Aix Marseille University, 2022  
 Laudine LÊ — Biometrics (M2), Paris-Est University, 2022  
 Inès Ben Haj Kacem — Cognitive Science (M1), Ecole Normale Supérieure, 2022  
 Sarah Said — Data Science (M1), Paris-Dauphine University, 2021  
 Gabriel Meunier — Artificial Intelligence (M2), Aix Marseille University, 2021  
 Kübra Bodur\* — Cognitive Science (M2), Aix Marseille University, 2021  
 Maureen Astier — Cognitive Science (M1), Aix Marseille University, 2021  
 Pelayo Mencos — Cognitive Science (M2), Aix Marseille University, 2020  
 Thomas Misiek\* — Cognitive Science, Aix Marseille University, 2020  
 Hang Jiang\* — Symbolic Systems (M1-M2), Stanford University, 2019

## Summer Interns

Martin Cuingnet — Ecole Normale Supérieure Paris-Saclay  
 Lucien Loriferne — Ecole Normale Supérieure Lyon  
 Petros Kaklamanis\* — University of Pennsylvania  
 Martin Padovani — Ecole Polytech' Marseille  
 Hong Duc Thang Vu\* — Aix Marseille University  
 Elias Semmar\* — Aix Marseille University  
 Cassandre Armand\* — Aix Marseille University  
 Charlie Hallart\* — Aix Marseille University  
 Morgane Peirollo\* — Aix Marseille University  
 Eliot Maes\* — Aix Marseille University  
 Yassine Laribi — Ecole Centrale Marseille  
 Zihan Xu\* — Jiao Tong University  
 Kyra Wilson\* — Stanford University  
 Yifan Ma — Stanford University  
 Sophie Regan\* — Stanford University  
 Isaac Scheinfeld\* — Stanford University

Yuan Bian\* — Stanford University

## Thesis Advisory Committee

Khazar Khorrami (Tampere University), May Abdou (Ecole Normale Supérieure), Kaijia Tey (Ecole Normale Supérieure), Rozic Gal (University College London), Rudradeep Guha (Université de Bourgogne), Daria Goryachun (Aix Marseille University), Maïwenn Fleig (Aix Marseille University), Ambre Balleroy (Aix Marseille University), Leonardo Pinto Arata (Aix Marseille University), Auriane Boudin (Aix Marseille), Alafate Abulimiti (INRIA Paris), Léo Jacqmin (Aix Marseille), Salma Ferjani (Aix Marseille).

## TEACHING

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### Aix Marseille University

#### **Computer Science Department**

Data Science (L1), 2025

Introduction to Machine Learning (L3), 2022–2023

Probability Theory for Computer Science (M1), 2022

Introduction to Natural Language Processing (M1), 2020–2023

Introduction to Computer Science (L1), 2020

Computer programming: Python (L1), 2020–2024

#### **Cognitive Science Master Program (MASCO)**

Building Minds: the Programming Tools (M1), 2025–present

Machine Learning for Cognitive Science (M2), with T. Schatz, 2024–present

Introduction to Cognitive Science (M1), 2020–2022

Introduction to Computer Programming (M1), 2020–2024

### Summer Schools

#### **ILCB summer school**

*Deep Learning for a theory of Language Development*, 2026 (week-long course)

*AI for the Ecological Study of Language Development*, 2025 (week-long course)

*Child language acquisition: State of the art, 2021–2024* (single-lecture course)

**ETAL (National Summer School on NLP)**

*When NLP meets Cognitive Science, 2023* (single-lecture course)

## SERVICE

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### Editorial Service

**Conference Editorial Roles**

*NLP Track: Linguistic Theories, Cognitive Modeling, and Psycholinguistics*

Senior Area Chair, COLING, 2025

Area Chair, ACL Rolling Review, 2025

Area Chair, ACL, 2023

**Associate Editor**, *Frontiers in Cognitive Neuroscience*, 2023–2025

**Guest Editor**, *Computational Linguistics*, 2024

Special issue: *Language Learning, Representation, and Processing in Humans and Machines*

Co-editors: M. Apidianaki (UPenn) and S. Padó (Stuttgart University)

**Guest Editor**, *Frontiers in Cognitive Neuroscience*, 2025

Special issue: *Neural and Behavioral Mechanisms of Social Learning*.

Co-editors: Marie H. Monfils (UT Austin) and Laura Agee (UT Austin)

### Reviewing Service

**Funding Agencies**

Agence Nationale de la Recherche (ANR, France)

National Science Foundation (NSF, USA),

The British Academy (UK)

The Royal Society (UK)

IVADO AI (Canada)

National Science Center (Poland)

**Book Proposals**

Cambridge University Press

**Journals (selection)**

Psychological Review, Trends in Cognitive Science, Proceedings of The Royal Society B, Cognition, Cognitive Science, Cognitive Psychology, Child Development, Journal of Experimental Psychology: General, Developmental Psychology, Developmental Cognitive Neuroscience, Open Mind: Discoveries in Cognitive Science, Speech Communication, Language Learning, Infant Behavior & Development, Language Learning & Development, Frontiers Human Neuroscience, Behavior Research Methods, Language & Cognition.

**Conferences (selection)**

ACL Rolling Review (ARR), Cognitive Science Society Conference (CogSci), International Conference on Multimodal Interaction (ICMI), Cognitive Development Society Conference (CDS), Conference on Computational Natural Language Learning (CoNLL), International Conference on Affective Computing and Intelligent Interaction (ACII), Workshop on Cognitive Modeling and Computational Linguistics (CMCL), Workshop on Knowledge Extraction and Integration for Deep Learning (DeeLIO).

## Governance &amp; Institutional Service

**Governing Board**, Institute of Language Communication & the Brain (ILCB), 2020–present

**Co-Founder and Director** (with L. Prévot), CoCoDev project-team, ILCB, 2020–present

**Co-Founder and Coordinator** (with P. Denis and L. Prévot), National Working Group (GT)

“*NLP & Cognition*” GDR TAL (CNRS), 2025 –present

**Co-Director** (with B. Isableu), Master in Cognitive Science (MaSCo), 2020–2021

**Selection Committee**, Maître de Conférences position in Computer Science, Univ. Lille, 2025

**Organizer**, The CoCoDev Seminars (archive and recordings), 2020–present

## Event co-organization

Workshop on *Computational Developmental Linguistics* (CDL), ACL, 2026

National Meeting of the GT “*NLP & Cognition*” GDR TAL (CNRS), 2025

Workshop on Multimodality in Social Interactions (MSI), 2024

National Summer School on NLP (ETAL), 2023

National meeting *Bridging NLP and Cognitive Science*, GDR TAL (CNRS), 2020

## Media & Public Outreach

**Interactive stand**, CoCoDev research, Festival Explore, 2026

**Interview** with Photizo on IA (Youtube podcast): *IA and Cognitive Science*, 2025

**Interactive stand**, the MACOMIC ANR project, Nuit des chercheurs, 2023

**Public outreach article**, “Les bébés apprennent l’art de la conversation avant même de savoir parler” in *The Conversation* ([link](#)), 2022